5.20 Solution: (a) By Eq. (5.12) and the definitions for the volume current and the surface current, the magnetic force is

$$\mathbf{F} = \int \left[(\nabla \times \mathbf{M}) \times \mathbf{B}\right] d^3x + \oint \left[(\mathbf{M} \times \mathbf{n}) \times \mathbf{B}\right] da.$$

Using the vector identities

$$\nabla(\mathbf{M} \cdot \mathbf{B}) = (\mathbf{M} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{M} + \mathbf{M} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{B}),$$

$$(\mathbf{M} \times \mathbf{n}) \times \mathbf{B} = \mathbf{n}(\mathbf{M} \cdot \mathbf{B}) - \mathbf{M}(\mathbf{n} \cdot \mathbf{B}),$$

and also the fact that $\nabla \times \mathbf{B} = 0$ in the absence of macroscopic current, we can express the force as

$$\mathbf{F} = \int \left[-\nabla(\mathbf{M} \cdot \mathbf{B}) + (\mathbf{M} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{M}\right] d^3x + \oint \left[\mathbf{n}(\mathbf{M} \cdot \mathbf{B}) - \mathbf{M}(\mathbf{n} \cdot \mathbf{B})\right] da$$

We can further simplify the result with some vector calculus theorems listed at the beginning of Jackson. First,

$$\int \nabla(\mathbf{M} \cdot \mathbf{B}) d^3x = \oint \mathbf{n}(\mathbf{M} \cdot \mathbf{B}) da.$$

Next, for well-behaved $\mathbf{a}$ and $\mathbf{b}$,

$$\begin{aligned}
\int (\mathbf{a} \cdot \nabla)\mathbf{b} &= \sum_{ij} \hat{e}_j \int (a_i \partial_i) b_j d^3x = \sum_{ij} \hat{e}_j \int \left[\partial_i(a_i b_j) - b_j(\partial_i a_i)\right] d^3x \\
&= \oint (\mathbf{a} \cdot \mathbf{n})\mathbf{b} da - \int (\nabla \cdot \mathbf{a})\mathbf{b} d^3x.
\end{aligned}$$

Therefore,

$$\int (\mathbf{M} \cdot \nabla)\mathbf{B} d^3x = \oint (\mathbf{M} \cdot \mathbf{n})\mathbf{B} da - \int (\nabla \cdot \mathbf{M})\mathbf{B} d^3x,$$

and

$$\int (\mathbf{B} \cdot \nabla)\mathbf{M} d^3x = \oint (\mathbf{B} \cdot \mathbf{n})\mathbf{M} da - \int (\nabla \cdot \mathbf{B})\mathbf{M} d^3x = \oint (\mathbf{B} \cdot \mathbf{n})\mathbf{M} da,$$

since $\nabla \cdot \mathbf{B} = 0$.

Putting everything together, we will have

$$\mathbf{F} = -\int (\nabla \cdot \mathbf{M})\mathbf{B} d^3x + \oint (\mathbf{M} \cdot \mathbf{n})\mathbf{B} da.$$

(b) Since the sphere has a uniform magnetization, $\nabla \cdot \mathbf{M} = 0$, and we only need to consider the surface integral term. The magnetization is in the (θ_0, ϕ_0) direction in spherical coordinates, then for a unit vector in the (θ, ϕ) direction,

$$\mathbf{M} \cdot \mathbf{n} = M \left[\cos\theta \cos\theta_0 + \sin\theta \sin\theta_0 \cos(\phi - \phi_0)\right].$$

Then, the surface integral becomes

$$\begin{aligned}
\mathbf{F} &= M B_0 R^2 \int \left[\cos\theta \cos\theta_0 + \sin\theta \sin\theta_0 \cos(\phi - \phi_0)\right] \begin{pmatrix} 1 + \beta R \sin\theta \sin\phi \\ 1 + \beta R \sin\theta \cos\phi \\ 0 \end{pmatrix} d\Omega \\
&= \frac{4}{3}\pi R^3 M \cdot \beta B_0 \begin{pmatrix} \sin\theta_0 \sin\phi_0 \\ \sin\theta_0 \cos\phi_0 \\ 0 \end{pmatrix}.
\end{aligned}$$